COMPASS: Convex Optimization of Mediated Polygenic Architecture via SNP Scores

Abstract

We present a unified methodological framework for estimating gene-specific mediated heritability from genome-wide association study (GWAS) summary statistics using gene-variant annotations. While Mediated Expression Score Regression (MESC) relies on observed eQTL effect sizes, COMPASS can leverage arbitrary gene-variant annotations (e.g., deep learning predictions, Hi-C contact frequencies). introduces an unknown annotation-weight vector, transforming the gene-level inference problem into a non-convex, bilinear trap. We demonstrate that relaxing the strict rank-1 assumption to a general gene-by-mechanism heritability matrix regularized by the nuclear norm yields a convex optimization landscape. This formulation natively identifies modular, multi-mechanism regulatory architectures across disease genes. Furthermore, we prove that enforcing strict non-negativity is essential to prevent linkage disequilibrium (LD) cancellation pathologies, where collinear LD among adjacent genes creates artificial, mutually cancelling positive and negative causal effect estimates. Finally, we provide an alternating projection algorithm combining Singular Value Thresholding (SVT) and non-negative clipping to fit the genome-wide model efficiently.

1 Model

Let ω_j denote the marginal effect of single nucleotide polymorphism (SNP) $j \in \{1, \ldots, J\}$ on a complex trait. We model ω_j as a causal cascade mediated through genes $k \in \{1, \ldots, K\}$:

$$\omega_j = \sum_{k=1}^{K} \beta_{jk}\alpha_k + \gamma_j \tag{1}$$

where β_{jk} is the true, unobserved regulatory effect of SNP j on gene k, α_k is the causal effect of gene k on the trait, and $\gamma_j \sim \mathcal{N}(0, \tau_\gamma)$ captures non-mediated (direct or pleiotropic) effects.

Instead of relying on empirical eQTLs, we utilize L non-negative annotations $a_{jkl} \geq 0$ representing the predicted regulatory potential of SNP j on gene k via biological mechanism $l \in \{1, \ldots, L\}$. We place a linear variance prior on the regulatory effects:

$$\beta_{jk} \sim \mathcal{N}(0, V_{jk}) \quad \text{where} \quad V_{jk} = \sum_{l=1}^{L} w_l a_{jkl} \tag{2}$$

Here, $w_l \geq 0$ represents the unknown heritability contribution weight of mechanism l.

1.1 The infinitesimal Gaussian assumption and its limitations

A simple assumption would be infinitesmal effects, $\alpha_k \sim \mathcal{N}(0, \tau_\alpha)$. Given that β_{jk} and α_k are independent and mean-zero, the variance of their product is the product of their variances:

$$\mathrm{Var}(\beta_{jk}\alpha_k) = \mathbb{E}[\beta_{jk}^2]\mathbb{E}[\alpha_k^2] = V_{jk}\tau_\alpha \tag{3}$$

Assuming independence across genes, we sum over genes, substitute the linear variance model for V_{jk}, and commute the sums:

$$\mathrm{Var}(\omega_j) = \sum_{k=1}^{K} V_{jk}\tau_\alpha + \tau_\gamma = \tau_\alpha \sum_{k=1}^{K}\left(\sum_{l=1}^{L} w_l a_{jkl}\right) + \tau_\gamma = \tau_\alpha \sum_{l=1}^{L} w_l \left(\sum_{k=1}^{K} a_{jkl}\right) + \tau_\gamma = \sum_{l=1}^{L} (\tau_\alpha w_l) A_{jl} + \tau_\gamma, \tag{4}$$

where $A_{jl} \equiv \sum_{k=1}^{K} a_{jkl}$. The gene-aggregated matrix A is therefore simply a $J \times L$ SNP annotation matrix: each column $A_{\cdot l}$ can be used as an ordinary S-LDSC annotation with coefficient $\theta_l = \tau_\alpha w_l$. Thus this model is almost trivial, involving just summing over genes.

1.2 Gene-specific resolution via conditional variance

To achieve gene-specific resolution we cannot rely on the infinitesimal Gaussian assumptions above. Here we consider conditioning the variance of the marginal effect on the specific causal trait variance of each gene, $s_k = \alpha_k^2 \geq 0$,

$$\mathrm{Var}(\omega_j \mid \alpha) = \sum_{k=1}^{K} s_k V_{jk} + \tau_\gamma \tag{5}$$

Substituting this conditional variance into the expected GWAS χ^2 statistic yields,

$$\begin{aligned}
\mathbb{E}[\chi_j^2 \mid \alpha] &= 1 + N \sum_{m=1}^{J} r_{jm}^2 \mathrm{Var}(\omega_m \mid \alpha) \\
&= 1 + N \sum_{m=1}^{J} r_{jm}^2 \left(\sum_{k=1}^{K} s_k \sum_{l=1}^{L} w_l a_{mkl} + \tau_\gamma\right) \\
&= 1 + N \sum_{k=1}^{K}\sum_{l=1}^{L} (s_k w_l)\mathcal{T}_{jkl} + N\tau_\gamma l_j
\end{aligned} \tag{6}$$

where r_{jm} is LD, $l_j = \sum_m r_{jm}^2$ is the standard non-mediated LD score, and the deterministic, pre-computable, sparse 3D annotation-gene LD Tensor,

$$\mathcal{T}_{jkl} = \sum_{m=1}^{J} r_{jm}^2 a_{mkl}. \tag{7}$$

Direct optimization of s and w has two pathologies:

1. **Scale Non-Identifiability:** For any scalar $c > 0$, the transformation $s \to s/c$ and $w \to cw$ leaves the product $B = sw^T$ unchanged, rendering the individual scales of gene importance and annotation importance unidentifiable without arbitrary anchor priors. This is addressable, for example, by constraining the L_1 norm of w (or s) to be 1, but this does not resolve the second pathology:

2. **Non-Convexity:** The bilinear optimization is non-convex, containing saddle points and local minima where optimization routines can get trapped.

1.3 Convex Nuclear Norm Relaxation

Let $B_{kl} = s_k w_l$ define the $K \times L$ matrix of mediated heritability contributions. We relax the strict rank-1 constraint $B = sw^T$ by treating $B \in \mathbb{R}^{K \times L}$ as a general matrix parameter regularized by the nuclear norm $\|B\|_* = \sum_i \sigma_i(B)$, where $\sigma_i(B)$ are the singular values of B.

We formulate the inference as a Weighted Ordinary Least Squares (WOLS) regression minimized over the space of non-negative matrices:

$$\min_{B \in \mathbb{R}_+^{K \times L},\, \tau_\gamma \geq 0} \quad \sum_{j=1}^{J} \eta_j \left(\chi_j^2 - 1 - N \sum_{k=1}^{K} \sum_{l=1}^{L} B_{kl} \mathcal{T}_{jkl} - N \tau_\gamma l_j \right)^2 + \lambda \|B\|_* \tag{8}$$

where η_j are standard S-LDSC regression weights accounting for LD redundancy and heteroskedasticity.

This relaxation yields three (potential) advantages:

- **Convexity:** Because the WOLS loss is convex with respect to B and the nuclear norm is the convex envelope of the rank function, the objective is convex, guaranteeing that any local minimum is the unique global optimum.

- **Discovery of Regulatory Modules:** Rather than enforcing regulatory homogeneity (where every causal gene obeys the exact same regulatory weight vector w), a low-rank solution $\text{rank}(B) = r > 1$ decomposes into r distinct mechanistic archetypes via Singular Value Decomposition (SVD):

$$B = \sum_{i=1}^{r} \sigma_i u_i v_i^T \tag{9}$$

  Here, the right singular vectors $v_i \in \mathbb{R}^L$ represent distinct biological pathways (e.g., promoter-driven vs. distal Hi-C loop enhancer-driven), while the left singular vectors $u_i \in \mathbb{R}^K$ cluster disease genes by their physical mechanism of action.

- **Computational Tractability:** Because the number of annotations L is much smaller than the number of genes ($K \approx 20,000$), computing the reduced SVD during optimization takes $O(KL^2)$ time.

2 Cell-Type Indexing

The index l need not correspond to a biochemical mechanism such as promoter activity, enhancer activity, or chromatin contact class. The same derivation applies if l indexes cell types or cellular contexts. In particular, consider the model

$$\beta_{jkl} \sim \mathcal{N}(0, w_l a_{jkl}) \tag{10}$$

where l indexes cell types. We now model,

$$\omega_j = \sum_{k=1}^{K} \sum_{l=1}^{L} \beta_{jkl} \alpha_{kl} + \gamma_j \tag{11}$$

where α_{kl} is the causal effect of gene k in cell type l on the trait. We then have,

$$\text{Var}(\omega_j | \alpha) = \sum_{k=1}^{K} \sum_{l=1}^{L} \alpha_{kl}^2 w_l a_{jkl} + \tau_\gamma = \sum_{k=1}^{K} \sum_{l=1}^{L} B_{kl} a_{jkl} + \tau_\gamma \tag{12}$$

where now $B_{kl} = \alpha_{kl}^2 w_l$ is the gene-by-cell-type mediated heritability matrix. The same convex nuclear norm relaxation applies, and the SVD of B now identifies cell-type-specific regulatory modules.

2.1 The Necessity of Non-Negativity

While standard summary statistic methods occasionally tolerate negative enrichment parameters, enforcing $B_{kl} \geq 0$ is mandatory in this gene-centric framework. Without non-negativity constraints, the optimization landscape suffers from two failures:

LD Collinearity Overfitting. Neighboring genes on a chromosome reside within the same LD block, causing their annotation tensors to be highly collinear ($\mathcal{T}_{j,k_1,l} \approx \mathcal{T}_{j,k_2,l}$ for adjacent genes k_1, k_2). If an unconstrained model attempts to fit a localized GWAS signal corresponding to a true heritability contribution of $+10$, it can overfit statistical noise by assigning extreme, mutually cancelling coefficients to adjacent genes, e.g. $B_{k_1,l} = +100$ and $B_{k_2,l} = -90$ so that $B_{k_1,l} + B_{k_2,l} = +10$. This "cancellation cheat" preserves the macroscopic model fit while destroying gene-level resolution, creating phantom causal effects. The constraint $B_{kl} \geq 0$ acts as a regularizer against LD collinearity, forcing the model to assign heritability exclusively to the most parsimonious gene.

Mathematical and Biological Plausibility By definition, $B_{kl} = \alpha_k^2 w_l$ represents the variance of the causal trait effect mediated by gene k via mechanism l. Variance cannot be negative. Further, because $\mathcal{T}_{jkl} \geq 0$, allowing negative entries in B permits the model to predict $\mathbb{E}[\chi_j^2] < 1$, implying that genetic variation suppresses sampling noise—a statistical impossibility under the null hypothesis.

2.2 Optimization via Alternating Projections

Because standard Singular Value Thresholding (SVT) introduces orthogonal vectors with negative values, we must solve a Non-Negative Low-Rank Matrix Approximation problem. We solve this efficiently using Proximal Gradient Descent with Dykstra's Alternating Projections.

Let $f(B, \tau_\gamma)$ denote the WOLS loss function from Equation (7). The algorithm proceeds as follows:

Algorithm 1 Non-Negative Low-Rank Gene Heritability Regression

Input: Pre-computed tensor $\mathcal{T} \in \mathbb{R}_+^{J\times K\times L}$, GWAS $\chi^2 \in \mathbb{R}^J$, LD scores $l \in \mathbb{R}^J$, step size ζ, regularization λ.
Initialize: $B^{(0)} \leftarrow \mathbf{0}_{K\times L}$, $\tau_\gamma^{(0)} \leftarrow 0$
while not converged **do**
  1. Gradient Step:
  $G^{(t)} \leftarrow B^{(t)} - \zeta\nabla_B f(B^{(t)}, \tau_\gamma^{(t)})$
  $\tau_\gamma^{(t+1)} \leftarrow \max\left(0,\ \tau_\gamma^{(t)} - \zeta\nabla_{\tau_\gamma} f(B^{(t)}, \tau_\gamma^{(t)})\right)$
  2. Singular Value Thresholding (Low-Rank Projection):
  Compute reduced SVD: $U, \Sigma, V^T = \mathrm{SVD}(G^{(t)})$
  $\tilde{\Sigma} \leftarrow \mathrm{diag}\left(\max(0,\ \sigma_i - \zeta\lambda)\right)$
  3. Non-Negative Projection:
  $B^{(t+1)} \leftarrow \max\left(\mathbf{0}_{K\times L},\ U\tilde{\Sigma}V^T\right)$
end while
Output: Modular heritability matrix $B^* = B^{(t+1)}$, background pleiotropy $\tau_\gamma^* = \tau_\gamma^{(t+1)}$.

This algorithm converges to the global optimal modular architecture, avoiding non-convexity and LD collinearity artifacts.

3 Application to Alzheimer Disease (AD) GWAS

Our default annotation source is the public Nasser et al. ABC prediction table: all element-gene connections with ABC score at least 0.015 across 131 biosamples. The ABC model predicts enhancer-gene regulation from enhancer activity and enhancer-promoter contact; the public table is distributed by the Engreitz Lab and is linked from their resource page. The ABC documentation defines the score as activity times contact, normalized over candidate elements within 5 Mb of the gene promoter, and notes that predictions are same-chromosome CRE-gene links.

For Alzheimer disease (AD), the default run uses the brain-labelled biosamples available in this table: `astrocyte-ENCODE`, `bipolar_neuron_from_iPSC-ENCODE`, and `H1_Derived_Neuronal_Progenitor_Culturec` Because the public table does not contain a microglia biosample, the default also includes myeloid proxy contexts: `CD14-positive_monocyte-ENCODE`, `CD14-positive_monocytes-Roadmap`, and `THP-1_macrophage-` The full 131-biosample table is downloaded and can be selected by passing `--abc-cell-types all`.

The implementation reads the ABC table in chunks, intersects each CRE interval with GWAS variants, and emits sparse variant-gene-context triples whose values are ABC scores. The regression row universe is not restricted to ABC-overlapping variants. Instead, COMPASS uses all GWAS variants represented in the UK Biobank LD reference panel. Variants outside ABC CREs have all-zero annotation rows but remain in the regression so they contribute to LD tagging and the residual non-ABC-mediated LD-score term. Nearby CREs for the same gene are collapsed into coarse LD-distance clusters before fold assignment.

LD is represented as chromosome-level sparse blocks rather than a single genome-wide matrix. Raw UK Biobank correlations are converted to r^2, entries with $r^2 < 0.01$ are discarded, and the retained LD values are stored and applied as half-precision sparse tensors during fitting. Other model tensors default to single precision, with a command-line option to compare half-precision model tensors against the default while keeping LD half precision in both runs.

3.1 Cross-validation strategy

Leaving out an entire chromosome is not a valid penalty-selection scheme for the gene-indexed COMPASS parameter B. If chromosome 1 is held out, then genes on chromosome 1 have no training annotations. Prediction on the held-out chromosome therefore relies on gene effects that were not estimable in the training set.

Our CV design therefore assigns fold labels to CRE groups rather than chromosomes. For each gene, CREs are sorted by genomic position after grouping nearby CREs into LD-distance clusters. Independent clusters for the same gene are assigned across folds. Variant-level fold labels are then inherited from the CRE groups they overlap; variants overlapping multiple conflicting folds are retained for training but are not used as validation rows. For a validation fold, the model is trained on all variants not assigned to that fold and evaluated on variants assigned to the held-out CRE fold.

This design has the intended target: the model learns B_{kl} from some CREs for a gene and tests whether those estimates predict summary statistics for variants in other, approximately LD-independent CREs for the same gene. It also preserves genome-wide LD modeling because the held-out units are local CRE clusters rather than whole chromosomes.

3.2 Implementation status

The repository now downloads the public ABC table into `raw/abc/AllPredictions.AvgHiC.ABC0.015.minus150.` and uses ABC annotations by default in `scripts/run.py`. Dataset caches store CRE fold labels alongside GWAS statistics, chromosome labels, and sample sizes. The row set is filtered to variants present in the UK Biobank LD metadata, and non-CRE rows are retained with zero annotations.